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Is the CS2 Skin Market Integrated with Traditional and Crypto Financial Markets? A Time Series Analysis

Time Series Analysis in Finance (TSA01_FS2601)

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Abstract

This paper examines whether the CS2 skin market — a multi-billion-dollar virtual goods market operating through the Steam Marketplace — is integrated with traditional and crypto financial markets. We construct a daily value-weighted price index from 1,607 liquid CS2 skins over the period August 2013 to June 2024 and test for market integration against the S&P 500, Bitcoin, and Gold using exploratory data analysis, GARCH(1,1) volatility modelling, and a VAR(4) model with Granger causality tests. We find no evidence of market integration: contemporaneous correlations between CS2 returns and all three comparison assets are indistinguishable from zero, and no Granger causal relationship is detected in either direction at the 5% significance level. CS2 exhibits extreme annualised volatility (114.56%) and GARCH persistence identical to Bitcoin (0.9990), but its price dynamics are driven by game-specific factors rather than macroeconomic forces. These results position CS2 skins as a genuinely segmented alternative asset class with potential diversification benefits for financial investors.

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List of Abbreviations

ACF	Autocorrelation Function
ADF	Augmented Dickey-Fuller (unit root test)
AIC	Akaike Information Criterion
BIC	Bayesian Information Criterion (Schwarz Criterion, SC)
CS2	Counter-Strike 2
FPE	Final Prediction Error
GARCH	Generalised Autoregressive Conditional Heteroskedasticity
HQ	Hannan-Quinn Information Criterion
PACF	Partial Autocorrelation Function
VAR	Vector Autoregression
VWAP	Volume-Weighted Average Price

Glossary

CS2 (Counter-Strike 2) Valve's tactical first-person shooter and the successor to Counter-Strike: Global Offensive (CS:GO). Launched in 2023, it operates the same Steam Marketplace ecosystem established in 2013, through which cosmetic items are traded at real-money prices.

Skin A purely cosmetic weapon or equipment modification in CS2 with no gameplay effect. Skins are bought, sold, and stored as digital assets on the Steam Marketplace and constitute the asset class studied in this paper.

StatTrak A skin variant equipped with an electronic counter that tracks the number of in-game kills. StatTrak skins command a persistent price premium relative to their non-StatTrak counterparts and are excluded from the index to avoid confounding price signals unrelated to market-wide demand.

Steam Marketplace Valve Corporation's open secondary market integrated into the Steam gaming platform, through which users buy and sell in-game items at prices set by supply and demand. Unlike closed in-game economies, the Steam Marketplace allows freely negotiated prices denominated in real currency, making it amenable to standard financial time series analysis.

Value-weighted index A price index in which each constituent's contribution to the index level is proportional to its trading volume. Formally, $I_t = \sum_i p_{it} v_{it} / \sum_i v_{it}$, where p_{it} is the volume-weighted average price of item i on day t and v_{it} is its trading volume. Analogous to market-capitalisation weighting in equity indices, this ensures that heavily traded items drive the index level.

Log return The continuously compounded return defined as $r_t = \ln(P_t/P_{t-1})$. Preferred over simple returns because log returns are approximately additive over time, yield a stationary series when prices are non-stationary, and are symmetric around zero for up and down moves of equal magnitude.

Granger causality A statistical concept introduced by Granger Granger (1969): variable X Granger-causes Y if past values of X improve forecasts of Y beyond what is achievable using Y's own history alone. Granger causality is a statement about predictability, not structural or physical causation.

Liquidity filter A selection criterion that excludes assets with insufficient trading activity from an index. In this paper, any base item recording fewer than two transactions in any calendar month is excluded, analogous to the SIX Swiss Exchange minimum turnover requirement for equity index inclusion SIX Swiss Exchange AG (2024).

Persistence ($\hat{\alpha} + \hat{\beta}$) In a GARCH(1,1) model, the sum of the ARCH coefficient $\hat{\alpha}$ and the GARCH coefficient $\hat{\beta}$. Values close to unity indicate slow mean-reversion of conditional variance: a volatility shock today affects expected variance far into the future. A value of exactly 1 defines an integrated GARCH (IGARCH) process.

1 Introduction

Counter-Strike 2 (CS2) is one of the most played competitive shooters in the world. Since 2013, players have been able to trade purely cosmetic weapon modifications — "skins" — through the Steam Marketplace at freely negotiated, real-money prices. Unlike the closed monetisation systems of competing titles, Steam operates an open secondary market, making CS2 skins an observable and quantifiable asset class. By early 2025, the ecosystem had grown to over 22,000 tradeable items with an estimated market value of $4.1 billion Guede-Fernández et al. (2025). Prior work has found attractive risk-adjusted returns and low correlation with traditional assets Dobrynskaya and Strelnikov (2025), but without applying a VAR or Granger causality framework; no study has formally tested market integration using these tools on a comprehensively constructed price index.

This paper fills that gap. We construct a daily value-weighted CS2 price index from raw Steam Marketplace transaction data and ask:

"Is the CS2 skin market integrated with traditional and crypto financial markets, and what is the direction and temporal dynamics of any such relationship?"

We test this against the S&P 500, Bitcoin, and Gold — representing traditional equity, speculative crypto, and safe-haven/inflation-hedge characteristics respectively — using EDA, GARCH(1,1) volatility modelling, and a VAR model with Granger causality tests.

2 Literature Review

The CS2 skin market has emerged as a candidate financial asset class. Operating through the Steam Marketplace — an open secondary market with real-currency pricing — it reached an estimated $4.1 billion in value by early 2025 Guede-Fernández et al. (2025). Unlike competing titles that restrict trading to developer-set prices, CS2's open market allows supply and demand to determine prices, making skins suitable for standard financial analysis Zhou (2024). Dobrynskaya and Strelnikov document average annual returns of 41.2% with a Sharpe ratio of 0.34 across nearly 5,000 CS skins from 2013 to 2024 — exceeding both the US equity market (0.25) and gold (0.12) Dobrynskaya and Strelnikov (2025). These findings motivate the central question of whether the CS2 market operates independently of, or co-moves with, mainstream financial assets.

Bitcoin serves as the primary speculative comparator. Baur et al. find that Bitcoin is held predominantly for speculative purposes, with returns uncorrelated with traditional asset classes under both normal and stressed market conditions Baur et al. (2018). Bouri et al. confirm that Bitcoin offers minimal hedging properties, with its portfolio value lying in diversification rather than risk reduction Bouri et al. (2017). The shared demographic between Bitcoin investors and CS2 skin traders — young, digitally native, and risk-tolerant — motivates the hypothesis that crypto market sentiment may Granger-cause skin prices. The most plausible transmission mechanism is a wealth channel: rising Bitcoin prices increase the disposable income of the crypto-adjacent gamer demographic, stimulating demand for CS2 skins and potentially driving prices upward with a short lag. Akin et al., using a VAR framework equivalent to the one applied here, find weak but statistically detectable interactions between Bitcoin, the S&P 500, and gold, establishing the empirical context against which CS2's behaviour is measured Akin et al. (2024).

The methodological toolkit draws on three established approaches. Vector Autoregression,

introduced by Sims, captures joint dynamics across multiple time series without imposing a-priori causal structure Sims (1980). Granger causality Granger (1969) operationalises directional predictability in a linear framework: X Granger-causes Y if past values of X improve forecasts of Y beyond what Y's own history provides. GARCH(1,1) Bollerslev (1986) models time-varying conditional variance and its clustering properties, enabling cross-asset comparison of shock persistence.

3 Methodology

3.1 Hypotheses

Three testable hypotheses are derived from the literature:

H1 CS2 skin index returns are significantly correlated with Bitcoin returns (market integration hypothesis).

H2 Bitcoin Granger-causes CS2 skin prices, but not vice versa — motivated by the shared speculative, crypto-adjacent user demographic.

H3 CS2 exhibits higher volatility than the S&P 500 but similar volatility clustering patterns to Bitcoin.

3.2 Data

CS2 skin transaction data are sourced from a publicly available Kaggle dataset (CC-BY-NC-SA 4.0) containing Steam Marketplace records for 22,446 item variants (10,833 distinct base items) spanning August 2013 to June 2024. The sample is restricted to wearable item categories (weapon skins, knives, gloves) and StatTrak variants are excluded. A liquidity filter drops any item recording fewer than two transactions in any calendar month SIX Swiss Exchange AG (2024), reducing the sample to 1,607 liquid base items. The daily CS2 index level is:

$$I_t = \frac{\sum_i p_{it} \cdot v_{it}}{\sum_i v_{it}} \tag{1}$$

where p_{it} is the item-level volume-weighted average price across wear conditions on day t and v_{it} is the corresponding total daily trading volume. This value-weighted aggregation mirrors market-capitalisation weighting in equity indices MSCI Inc. (2024), ensuring that heavily traded skins drive the index level. The index spans 3,960 trading days. Daily log returns for the S&P 500 (`^GSPC`), Bitcoin (`BTC-USD`), and Gold (`GC=F`) are retrieved via `quantmod`. The four series are aligned via inner join on common trading days, yielding 2,448 observations (September 2014 to June 2024) used in all multivariate analyses.

3.3 Analytical Pipeline

Exploratory data analysis covers descriptive statistics, a contemporaneous Pearson correlation matrix (H1), and Augmented Dickey-Fuller tests with BIC lag selection and a constant term to confirm stationarity of log returns. A GARCH(1,1) model with student-t innovations is then fitted to each return series to compare unconditional volatility and shock persistence across assets (H3). Finally, a VAR(p) model is estimated on the joint

system with the lag order selected by BIC; Granger causality is tested via F-tests on jointly excluded lag coefficients in both directions between each benchmark and CS2 (H2).

4 Results and Discussion

4.1 Exploratory Data Analysis

Table 1 reports descriptive statistics for all four return series over the aligned sample of 2,448 observations. CS2 is the most volatile asset at 114.56% annualised, exceeding Bitcoin (69.87%), consistent with the first part of H3. The correlation matrix (Table 2) shows near-zero contemporaneous correlations between CS2 and all three benchmarks ($r_{\text{CS2,BTC}} = -0.016$, $r_{\text{CS2,SP500}} = 0.000$, $r_{\text{CS2,Gold}} = 0.011$), constituting direct evidence against H1. ADF tests reject the unit root null at the 1% level for all series, confirming that log returns are stationary and no further differencing is required.

Table 1: Descriptive Statistics — Daily Log Returns

Asset	Mean (%)	Std Dev (%)	Skewness	Ex. Kurtosis	Ann. Vol. (%)
CS2	0.009	7.217	−0.37	3.82	114.56
S&P 500	0.041	1.133	−0.80	15.74	17.98
Bitcoin	0.203	4.401	−0.63	9.07	69.87
Gold	0.026	0.927	−0.06	3.77	14.71

Table 2: Correlation Matrix — Daily Log Returns

	CS2	S&P 500	Bitcoin	Gold
CS2	1.000	0.000	−0.016	0.011
S&P 500	0.000	1.000	0.213	0.020
Bitcoin	−0.016	0.213	1.000	0.084
Gold	0.011	0.020	0.084	1.000

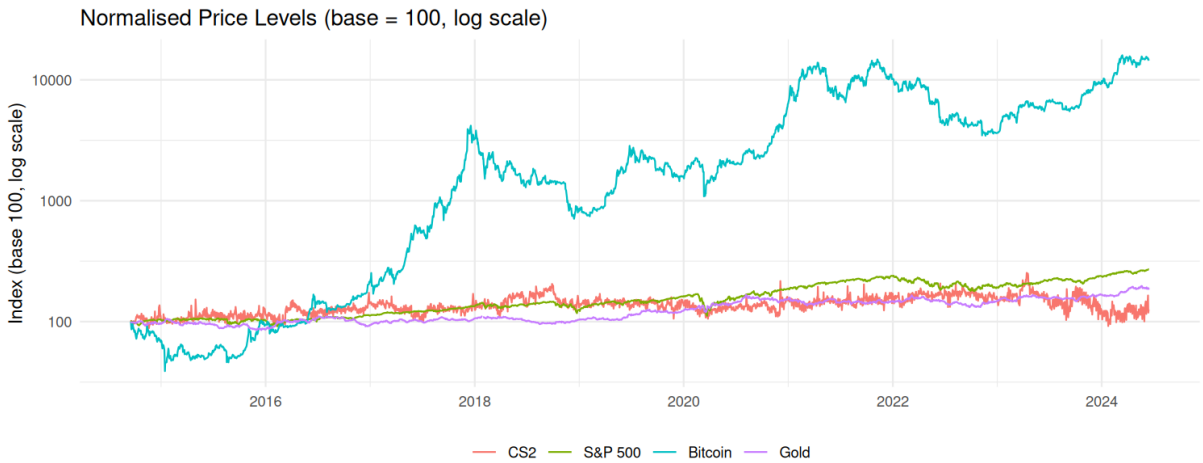

Figure 1: Normalised price levels — CS2 index, S&P 500, Bitcoin, and Gold (2013–2024, rebased to 100 at start of common sample; log scale).

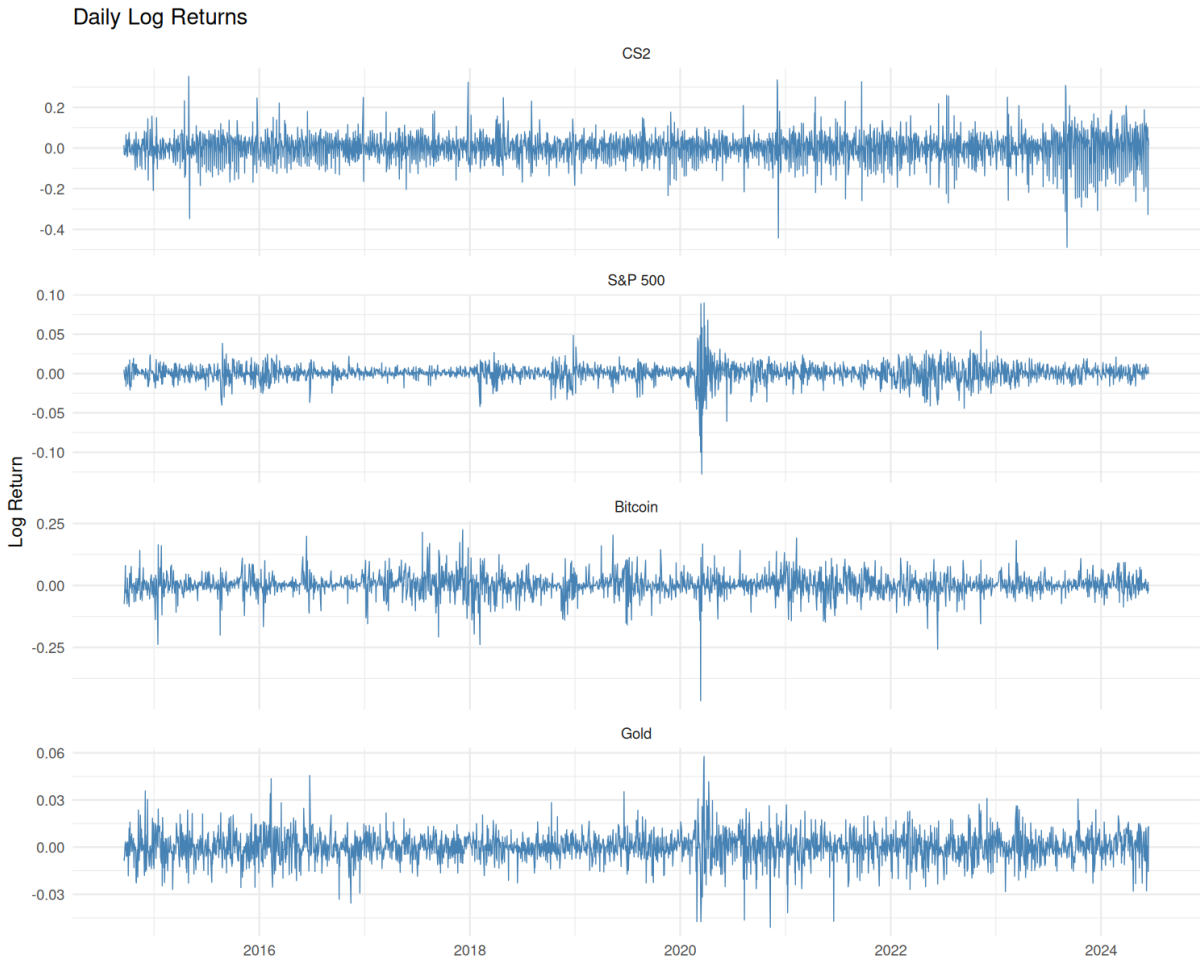

Figure 2: Daily log returns — CS2 index, S&P 500, Bitcoin, and Gold (2013–2024). CS2 and Bitcoin exhibit substantially wider return swings than the S&P 500 and Gold throughout the sample.

4.2 GARCH Volatility Comparison (H3)

Table 3 reports estimated GARCH(1,1) parameters. H3 is *partially supported*. CS2 unconditional volatility (161.88%) more than triples the S&P 500 (49.70%), confirming the first part. The unconditional GARCH volatility exceeds the sample standard deviation (114.56%) because near-integrated variance ($\hat{\alpha}+\hat{\beta} = 0.9990$) implies a high model-implied long-run variance that the finite sample mean only partially captures. Turning to volatility dynamics, CS2 and Bitcoin share identical total persistence ($\hat{\alpha}+\hat{\beta} = 0.9990$), meaning shocks to both markets decay at the same long-run rate. However, the decomposition reveals fundamentally different intraday mechanisms: CS2's ARCH coefficient ($\hat{\alpha} = 0.0151$) is an order of magnitude below Bitcoin's (0.1265), meaning CS2 volatility barely reacts to individual return shocks but instead persists in long-duration regimes driven by the GARCH term ($\hat{\beta} = 0.9839$). Bitcoin, by contrast, spikes sharply after each shock and decays from a higher peak. This distinction — equal long-run persistence but qualitatively different shock-response profiles — is consistent with the Steam Marketplace's closed trading hours, illiquid order book, and slow information transmission, which suppress short-term reactivity while allowing elevated variance states to persist.

Table 3: GARCH(1,1) Estimates

Asset	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\alpha}+\hat{\beta}$	Uncond. Vol. (%)
CS2	0.0151	0.9839	0.9990	161.88
S&P 500	0.1821	0.8158	0.9979	49.70
Bitcoin	0.1265	0.8725	0.9990	337.76
Gold	0.0280	0.9642	0.9923	16.61

4.3 VAR and Granger Causality (H1, H2)

A VAR(4) model is estimated on the joint system (BIC criterion; AIC suggests $p = 9$). Table 4 reports Granger causality F-tests; each row tests whether the cause variable Granger-causes the *entire remaining system jointly.*

Table 4: Granger Causality Tests — VAR(4)

Cause variable	F-statistic	p-value	Conclusion
Bitcoin	1.453	0.134	Not significant
S&P 500	1.275	0.226	Not significant
Gold	1.353	0.181	Not significant
CS2	1.663	0.068	Not significant (borderline)

H1 is **rejected** on the basis of the correlation matrix (Table 2): contemporaneous co-movement with all three benchmarks is indistinguishable from zero. H2 is **rejected** by the Granger tests: Bitcoin does not predict the joint system at any conventional significance level ($p = 0.134$), and the reverse direction is also non-significant. The borderline CS2 signal ($p = 0.068$) is noteworthy: while below the 5% threshold, it runs in the opposite direction to H2, suggesting that if any linear predictive link exists, it flows *from* CS2 to the broader system rather than towards it. Given that the test is joint, it is not possible to attribute this signal to any single benchmark series. Together, the absence of contemporaneous correlation and dynamic Granger predictability position CS2 as a genuinely segmented market whose price dynamics are driven by game-specific factors — patch updates, tournament cycles, case releases — rather than by macroeconomic or crypto forces.

4.4 Critical Discussion

The most direct explanation for the low co-movement is informational isolation: CS2 skin prices respond to game-specific news — case releases, patch updates, tournament activity, and developer announcements — rather than to macroeconomic or crypto signals. The Steam Marketplace's 15% transaction fee reinforces this by making arbitrage between the skin market and other asset classes prohibitively costly, while the inability to short-sell skins prevents price-correcting flows from external markets. Ljung-Box tests on VAR(4) residuals (Table 7) reveal significant residual autocorrelation in the CS2 and S&P 500 equations ($Q(10) > 90$, $p < 0.001$), indicating that VAR(4) does not fully capture CS2's dynamics. However, this misspecification biases toward conservative inference: residual autocorrelation inflates standard errors and makes it harder to reject the null of no Granger causality. A correctly specified, higher-lag model would therefore be even less

likely to find significance. The value-weighted index concentrates on high-volume skins, suppressing mean returns relative to an equal-weighted benchmark and partly explaining the low mean return (0.009%/day) relative to Dobrynskaya and Strelnikov's 41% annual figure Dobrynskaya and Strelnikov (2025). The September 2023 CS2 engine update may have introduced a structural break not captured by the constant-parameter models. Finally, the joint Granger specification is conservative; a pairwise CS2–Bitcoin VAR would provide a more focused robustness check for H2.

5 Conclusion

This paper investigated whether the CS2 skin market is integrated with traditional and crypto financial markets. Using a daily value-weighted price index from 1,607 liquid skins over the period August 2013 to June 2024, we tested for integration against the S&P 500, Bitcoin, and Gold via EDA, GARCH(1,1), and a VAR(4) with Granger causality tests.

The evidence consistently points to a lack of integration: the CS2 skin market is not integrated with any of the three comparison assets. Contemporaneous correlations are indistinguishable from zero and no Granger causal relationship is detected at the 5% significance level in either direction. H1 and H2 are rejected; H3 is partially supported — CS2 shares Bitcoin's total volatility persistence ($\hat{\alpha} + \hat{\beta} = 0.9990$) but differs markedly in shock reactivity, with CS2 exhibiting regime-like plateaus rather than Bitcoin's spike-and-decay pattern. The near-zero correlation with all mainstream assets confers meaningful diversification potential for investors already holding equity or crypto positions, corroborating the low-correlation finding of Dobrynskaya and Strelnikov Dobrynskaya and Strelnikov (2025). However, CS2 skins function as an idiosyncratic zero-beta diversifier rather than a systematic hedge: their value lies in low correlation with other markets, not in negative co-movement. Extreme annualised volatility (114.56%) and the absence of price-discovery linkages to broader markets further limit their practical appeal for risk-averse investors.

The main limitations are the restriction to Steam Marketplace data (excluding third-party platforms such as Skinport or CS.Money), the value-weighting bias towards liquid skins, and the potential structural break at the CS:GO-to-CS2 engine transition in September 2023. Future work should test for this breakpoint explicitly, estimate a pairwise CS2–Bitcoin VAR as a robustness check, and incorporate data from third-party platforms and equal-weighted benchmarks to obtain a more complete picture of CS2 skin price dynamics.

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A Appendix

A.1 Full EDA Plots

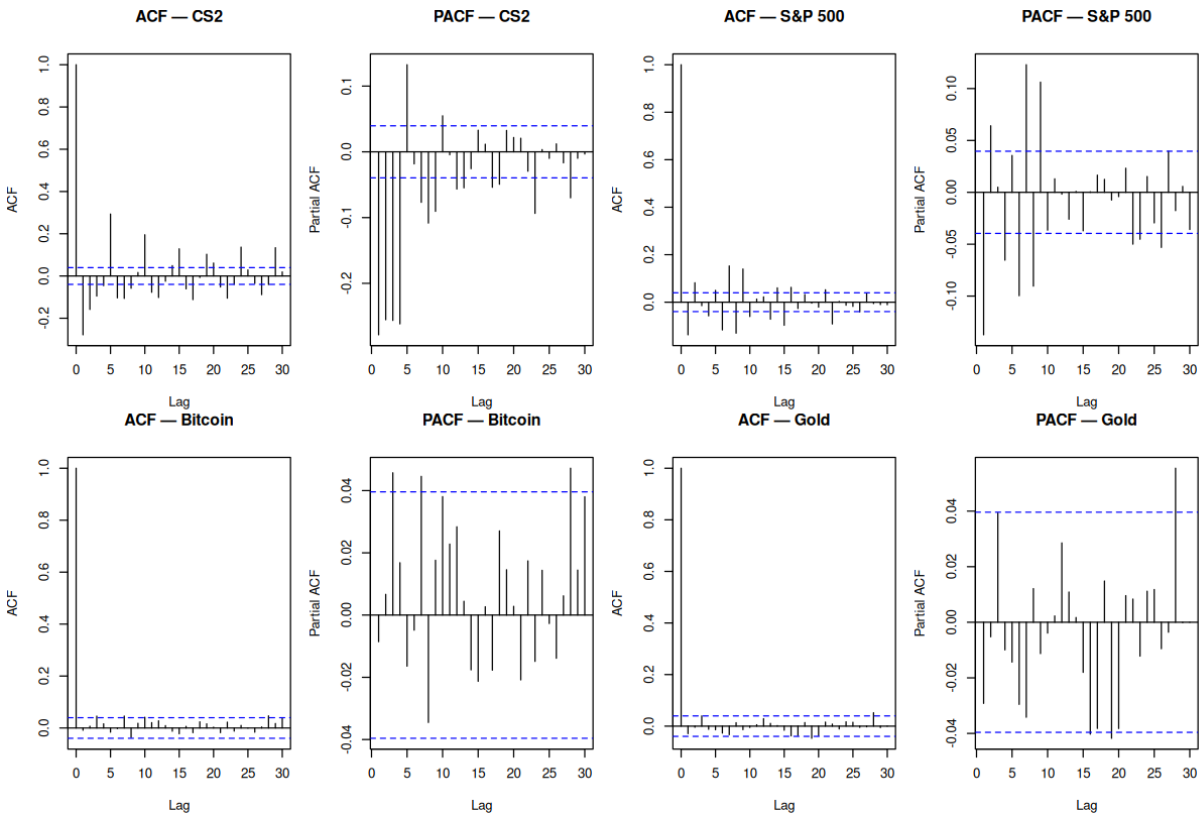

Figure 3: Autocorrelation (ACF) and Partial Autocorrelation (PACF) of daily log returns for all four assets at up to 30 lags.

The ACF plots confirm two key properties of the return series. First, the ACF of log returns shows no significant linear autocorrelation at any lag for any asset — a necessary condition for the absence of exploitable linear predictability in returns. (Note: the VAR(4) residuals for CS2 do reveal remaining structure not visible in the raw ACF; see Table 7 for details.) Second, the ACF of squared returns reveals significant, slowly decaying autocorrelation for all assets, a hallmark of ARCH effects and the empirical motivation for the GARCH specification. CS2 squared returns exhibit noticeably slower decay than Bitcoin and the S&P 500, indicating that CS2 volatility shocks persist in long regimes rather than dissipating quickly — consistent with the low ARCH coefficient ($\hat{\alpha} = 0.0151$) and high GARCH coefficient ($\hat{\beta} = 0.9839$) estimated in Section 4.2.

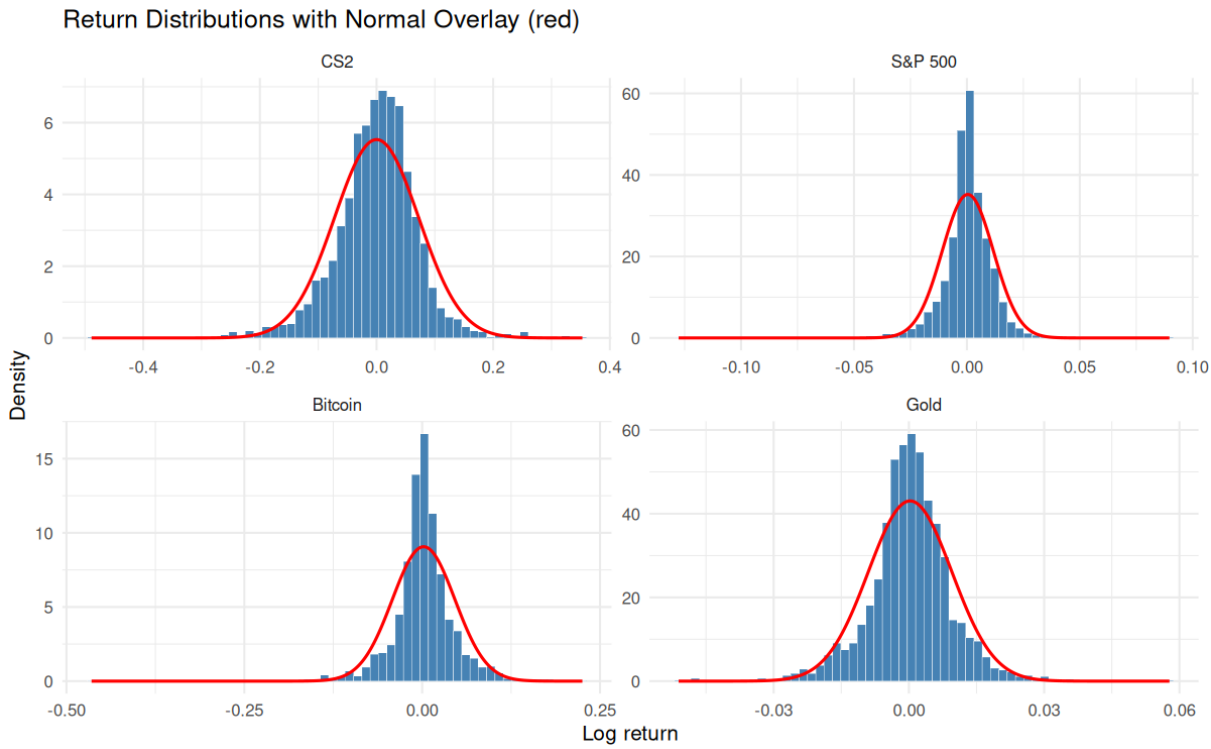

Figure 4: Empirical return distributions (histogram) overlaid with a fitted normal curve (red) for all four assets.

All four return series are leptokurtic, exhibiting fatter tails than the normal distribution. The departure is most pronounced for the S&P 500 (excess kurtosis 15.74) and Bitcoin (9.07), reflecting the high frequency of extreme return events in both markets. CS2 shows moderate excess kurtosis (3.82), above that of Gold (3.77), but below the two more liquid markets — plausibly because the Steam Marketplace's slow price-adjustment microstructure dampens extreme one-day moves. The systematic fat-tail pattern across all assets confirms the appropriateness of student-t innovations in the GARCH specification (rather than a Gaussian assumption), and the absence of a pronounced second mode in any distribution rules out bimodality as a confound.

A.2 Index Construction Details

Table 5: Item Filter Funnel — CS2 Index Construction

Stage	Item variants	Base items
Raw items (all categories, all variants)	22,446	10,833
After wearable + StatTrak/Souvenir filter	7,348	1,638
After liquidity filter ($\geq$2 trades/month)	—	1,607

Table 5 documents the item reduction at each construction stage. The raw dataset comprises 22,446 distinct item variants (combinations of base item and wear condition) across 10,833 base items spanning all Steam Marketplace categories. Restricting to wearable categories (weapon skins, knives, gloves) and excluding StatTrak and Souvenir variants reduces the sample to 7,348 item variants across 1,638 base items. This step removes

items whose prices are inflated by premium mechanics rather than driven by the underlying cosmetic market, ensuring the index reflects comparable, freely-tradeable assets. The final liquidity filter — requiring at least two recorded transactions in every calendar month — removes a further 31 base items (2%), retaining 1,607 items. The criterion is analogous to the SIX minimum turnover requirement for equity index inclusion SIX Swiss Exchange AG (2024): items failing it trade too infrequently for their daily prices to be reliable signals, and including them would introduce stale-price bias into the index level.

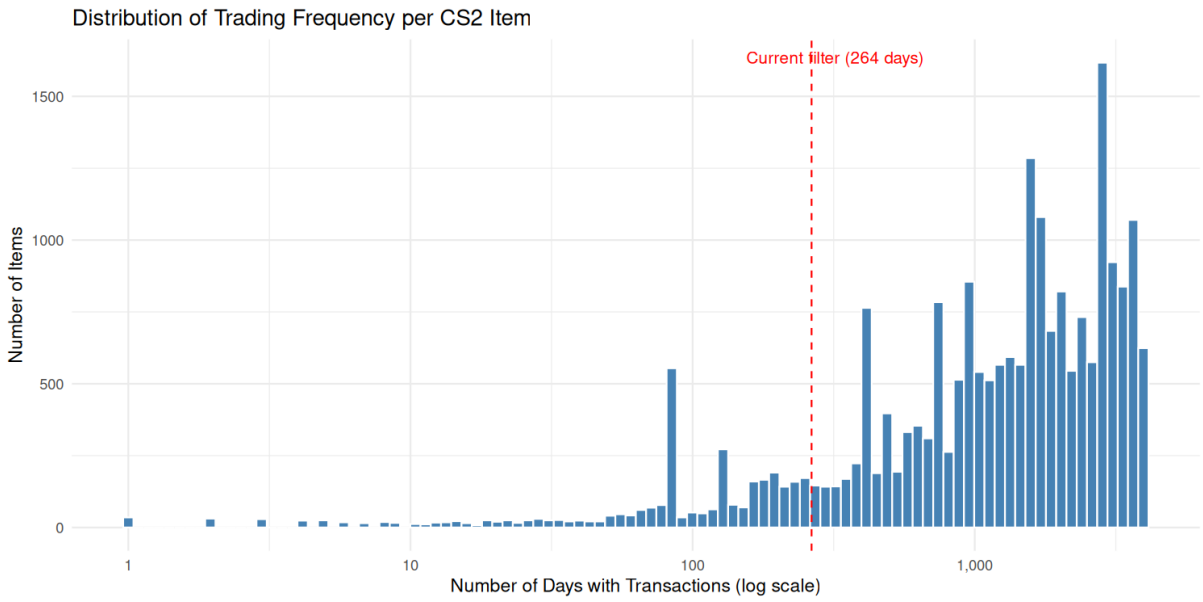

Figure 5: Distribution of trading frequency per CS2 item (number of days with at least one recorded transaction).

Figure 5 shows the distribution of trading days per item across the full raw dataset (log scale). The distribution is strongly right-skewed: a large mass of items trade on very few days and a smaller group of liquid items trade consistently throughout the sample. The vertical dashed line marks the current liquidity threshold: items to its left are excluded because at least one calendar month in their history contains fewer than two transactions. The retained items (to the right) contribute stable daily price observations throughout the 3,960-day sample, supporting the reliability of the value-weighted index.

A.3 GARCH Diagnostics

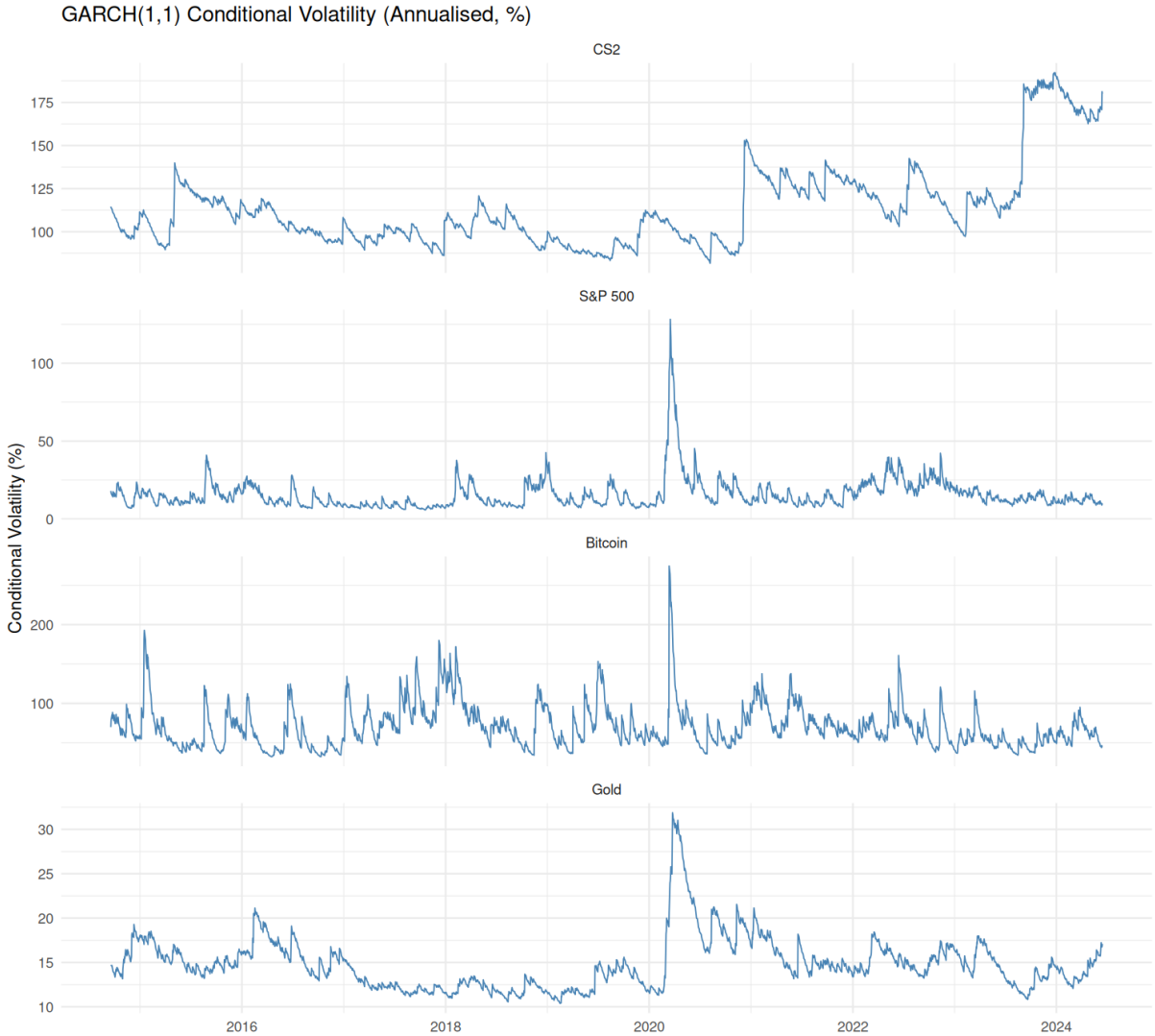

Figure 6: GARCH(1,1) conditional volatility (annualised, %) for all four assets, August 2014 to June 2024.

Figure 6 shows the estimated conditional volatility series for each asset. Several well-known market stress episodes are visible in the comparison assets: the S&P 500 spike in March 2020 (COVID-19 shock), the Bitcoin collapse of November 2022 (FTX bankruptcy), and the gold spike in March 2022 (Russia–Ukraine war). CS2 exhibits a qualitatively different profile: elevated volatility tends to persist in multi-month regimes rather than arriving as discrete spikes. This is consistent with the ARCH coefficient $\hat{\alpha} = 0.0151$ (far below Bitcoin's 0.1265), which means individual return shocks transmit only weakly into the next day's conditional variance, and the GARCH coefficient $\hat{\beta} = 0.9839$, which ensures that whatever variance level is reached decays extremely slowly. The contrast with Bitcoin — which has the same persistence (0.9990) but reacts sharply to individual shocks — confirms that the Steam Marketplace's closed trading hours, illiquid order book, and slow information transmission shape a distinct volatility microstructure.

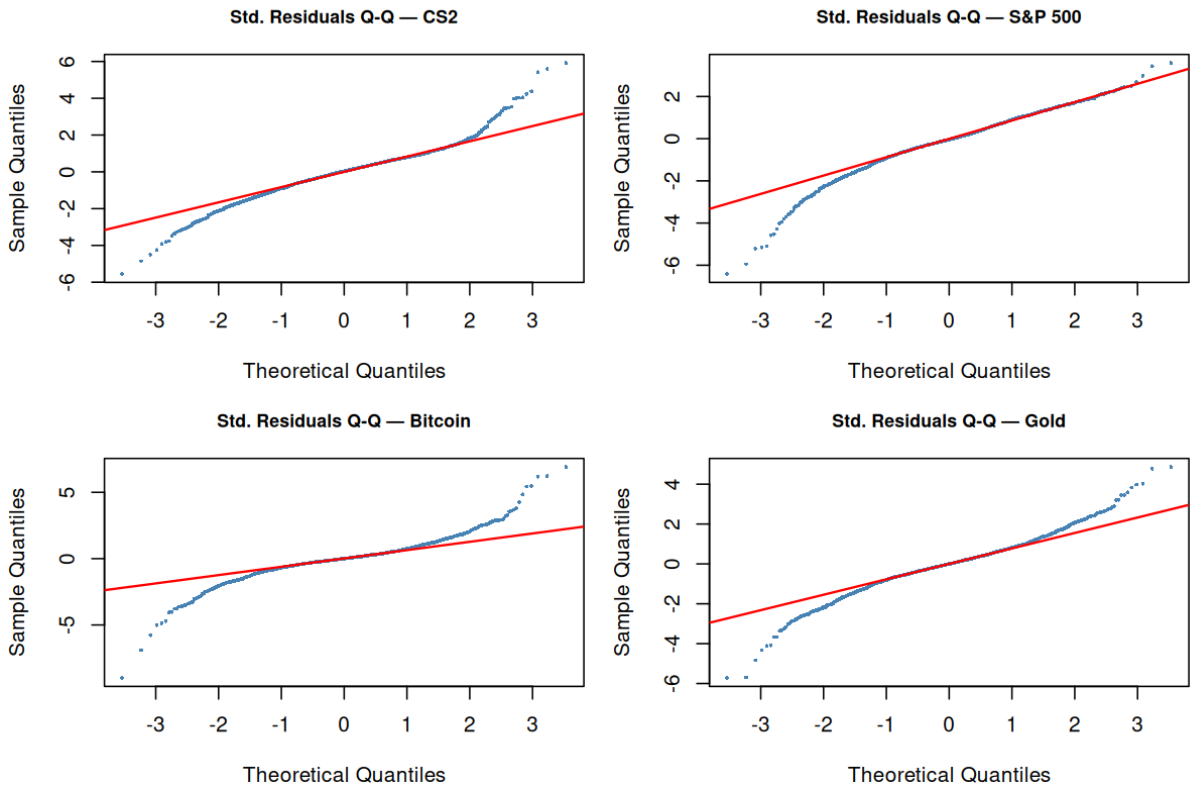

Figure 7: Q-Q plots of standardised GARCH(1,1) residuals against the standard normal for all four assets. Systematic tail departures from the diagonal motivate the student-t innovation assumption.

Figure 7 plots standardised GARCH(1,1) residuals against the standard normal quantiles. The plots are not an adequacy check against the fitted student-t distribution; rather, they serve as a diagnostic showing *why* the Gaussian assumption is inappropriate and why the student-t specification was chosen. All four assets exhibit systematic heavy-tail departures (points fan above the upper diagonal and below the lower diagonal), confirming the fat-tailed nature of standardised residuals. The central mass of each distribution tracks closely along the diagonal, indicating that GARCH(1,1) successfully removes the conditional mean structure and that the residual non-normality is concentrated in the tails, precisely where the student-t distribution provides additional flexibility relative to the Gaussian.

A.4 VAR Diagnostics

Table 6: VAR Lag Selection Criteria for Lags 1–10 (lower is better)

Lag	AIC	HQ	BIC	FPE ($\times 10^{-13}$)
1	−29.969	−29.951	−29.921	0.965
2	−30.037	−30.006	−29.952	0.901
3	−30.099	−30.054	−29.976	0.847
4	**−30.172**	**−30.114**	**−30.011**	**0.788**
5	−30.184	−30.111	−29.984	0.778
6	−30.190	−30.103	−29.952	0.774
7	−30.207	−30.107	−29.931	0.761
8	−30.222	−30.108	−29.908	0.749
9	−30.234	−30.106	−29.882	0.741
10	−30.231	−30.089	−29.840	0.743

Bold row: BIC optimum (p = 4). AIC and FPE optimum: p = 9.

Table 6 reports information criteria for lag orders 1–10. BIC (labelled SC in the `vars` package) selects $p = 4$, while AIC and FPE select $p = 9$. The penalty differential between AIC and BIC grows with the number of parameters, making BIC more conservative in models with a large endogenous dimension — here, 4 variables $\times$ p lags per equation. The BIC-selected VAR(4) uses $4 \times 4 \times 4 + 4 = 68$ parameters, which is already a substantial parameterisation for $N = 2{,}444$ effective observations. The marginal improvement in AIC from lag 4 to lag 9 is modest (from −30.172 to −30.234), suggesting that the additional lags capture only weak dynamics and that the parsimonious BIC choice is appropriate for inference.

Table 7: Ljung-Box Residual Autocorrelation Tests — VAR(4), Lag 10

Asset	$Q(10)$	p-value	Conclusion
CS2	92.11	< 0.001	Residual autocorrelation detected
S&P 500	98.26	< 0.001	Residual autocorrelation detected
Bitcoin	10.36	0.410	No significant autocorrelation
Gold	5.78	0.833	No significant autocorrelation

Table 7 reports univariate Ljung-Box tests on VAR(4) residuals. Bitcoin and Gold show no evidence of residual autocorrelation at lag 10 ($p > 0.4$), confirming that VAR(4) is adequate for those series. CS2 and the S&P 500 exhibit significant residual autocorrelation ($p < 0.001$), indicating that four lags are insufficient to fully capture the joint dynamics. For CS2, this is unsurprising given the strong negative first-order autocorrelation in the raw series (AR coefficient at lag 1: $\hat{\phi}_{11} = -0.485$, $p < 0.001$) and the Steam Marketplace's two-day settlement lag, which imparts a mechanical negative autocorrelation. For the S&P 500, weekly seasonality in trading patterns likely contributes.

These findings have two implications for the main analysis. First, the AIC's preferred order of $p = 9$ would reduce residual autocorrelation but at the cost of 80 additional parameters, raising degrees-of-freedom concerns and multicollinearity risks with $N = 2{,}444$ effective observations. Second, the Granger causality conclusions are not materially

affected: the tests found no significant predictive power from any benchmark to CS2 or vice versa, and residual misspecification of this type generally inflates standard errors, making it harder to reject the null of no causality — if anything, a better-specified model would be even less likely to find significance. This argument addresses test size rather than coefficient bias; a higher-lag VAR or an ARMA-augmented specification for CS2 would be needed to fully resolve the residual autocorrelation, and constitutes a natural direction for future work.

A.5 R Code

All analysis is implemented in R using `renv` for reproducible package management. The pipeline runs sequentially through five scripts; each section below documents one step. Package dependencies are declared at the top of each script and installed via `renv::restore()`.

Step 1 — Data Preprocessing (`src/01_data_preprocessing.R`)

Reads all raw item CSV files exported from the Steam Marketplace, decodes URL-encoded filenames using a name conversion table, parses structured metadata (weapon, skin name, wear condition, item category, StatTrak/Souvenir flags), aggregates intraday records to daily VWAP, and saves the result to `data/processed/cs2_daily.parquet`.

Listing 1: Step 1: Data preprocessing and daily VWAP aggregation

```r
library(data.table); library(arrow); library(here); library(stringr)

ITEMS_DIR  <- here("data", "raw", "items")
NAME_TABLE <- here("data", "raw", "name_conversion_table.csv")
OUT_FILE   <- here("data", "processed", "cs2_daily.parquet")
BATCH_SIZE <- 1000

WEAR_CONDITIONS <- c(
  "Factory New", "Minimal Wear", "Field-Tested", "Well-Worn", "Battle-Scarred"
)

# Name lookup table (URL-encoded -> decoded item names)
name_tbl    <- fread(NAME_TABLE, encoding = "UTF-8")
setnames(name_tbl, c("encoded", "decoded"))
name_lookup <- setNames(name_tbl$decoded, name_tbl$encoded)

# Parse structured metadata from a decoded item name
parse_item_name <- function(decoded_name) {
  name        <- decoded_name
  is_special  <- startsWith(name, "★ ")
  is_stattrak <- grepl("StatTrak ", name, fixed = TRUE)
  is_souvenir <- grepl("Souvenir",  name, fixed = TRUE)

  name <- sub("^★ \\s*", "", name)
  name <- sub("StatTrak \\s*", "", name)
  name <- sub("Souvenir\\s*", "", name)
  name <- trimws(name)

  wear <- NA_character_
  for (w in WEAR_CONDITIONS) {
    pattern <- paste0("\\s*\\(", w, "\\)$")
```

```r
    if (grepl(pattern, name)) {
      wear <- w; name <- trimws(sub(pattern, "", name)); break
    }
  }

  parts     <- strsplit(name, "␣|␣", fixed = TRUE)[[1]]
  weapon    <- trimws(parts[1])
  skin_name <- if (length(parts) >= 2) trimws(paste(parts[-1], collapse=
      "␣|␣"))
               else NA_character_

  item_category <- dplyr::case_when(
    grepl("Graffiti",  weapon,       fixed=TRUE) ~ "graffiti",
    grepl("Music␣Kit", decoded_name, fixed=TRUE) ~ "music_kit",
    grepl("Capsule",   decoded_name, fixed=TRUE) ~ "capsule",
    grepl("Case",      weapon,       fixed=TRUE) ~ "case",
    grepl("Package",   decoded_name, fixed=TRUE) ~ "case",
    grepl("Sticker",   weapon,       fixed=TRUE) ~ "sticker",
    is_special & grepl("Gloves|Wraps|Hand", weapon) ~ "glove",
    is_special                                      ~ "knife",
    !is.na(wear)                                    ~ "weapon_skin",
    TRUE                                            ~ "other"
  )

  base_item <- if (!is.na(skin_name)) paste0(weapon, "␣|␣", skin_name)
      else weapon
  list(weapon=weapon, skin_name=skin_name, wear=wear, base_item=base_
      item,
       item_category=item_category, is_stattrak=is_stattrak,
       is_special=is_special, is_souvenir=is_souvenir)
}

# Process one CSV file: decode name, aggregate to daily VWAP, attach
    metadata
process_file <- function(filepath) {
  enc_name <- tools::file_path_sans_ext(basename(filepath))
  dec_name <- name_lookup[enc_name]
  if (is.na(dec_name)) dec_name <- URLdecode(enc_name)

  dt <- tryCatch(fread(filepath, showProgress=FALSE, select=1:4),
                 error=function(e) NULL)
  if (is.null(dt) || nrow(dt) == 0) return(NULL)
  setnames(dt, c("price", "quantity", "date_str", "unix_ts"))

  dt[, date := as.Date(as.POSIXct(unix_ts, origin="1970-01-01", tz="UTC"
      ))]
  dt <- dt[, .(price    = sum(price * quantity) / sum(quantity),
               quantity = as.integer(sum(quantity))), by = date]

  meta <- parse_item_name(dec_name)
  dt[, `:=`(item_name=dec_name, item_category=meta$item_category,
            weapon=meta$weapon, skin_name=meta$skin_name, wear=meta$
                wear,
            base_item=meta$base_item, is_stattrak=meta$is_stattrak,
            is_special=meta$is_special, is_souvenir=meta$is_souvenir)]
  dt
}

```

```r
# Batched processing over all CSV files
files     <- list.files(ITEMS_DIR, pattern="\\.csv$", full.names=TRUE)
n_batches <- ceiling(length(files) / BATCH_SIZE)
batches   <- vector("list", n_batches)

for (b in seq_len(n_batches)) {
  idx        <- ((b-1)*BATCH_SIZE+1):min(b*BATCH_SIZE, length(files))
  batch      <- lapply(files[idx], process_file)
  batches[[b]] <- rbindlist(Filter(Negate(is.null), batch), fill=TRUE)
}

cs2 <- rbindlist(batches, fill=TRUE)
cs2[, trading_value := price * quantity]
setcolorder(cs2, c("date","item_name","item_category","weapon","skin_name",
                   "wear","base_item","is_stattrak","is_special","is_souvenir",
                   "price","quantity","trading_value"))
setorder(cs2, item_name, date)

write_parquet(cs2, OUT_FILE)
```

Step 2 — Index Construction (`src/02_index_construction.R`)

Filters to wearable item categories, excludes StatTrak and Souvenir variants, applies the monthly liquidity filter, and aggregates surviving items into a daily value-weighted index. Saves the index to `data/processed/cs2_index.parquet` and a filter funnel summary to `data/processed/filter_stats.csv`.

Listing 2: Step 2: Liquidity filter and value-weighted index construction

```r
library(data.table); library(arrow); library(here)

IN_FILE  <- here("data", "processed", "cs2_daily.parquet")
OUT_FILE <- here("data", "processed", "cs2_index.parquet")
WEARABLE_CATS <- c("weapon_skin", "knife", "glove")
LIQUIDITY_MIN <- 2L   # minimum trades per calendar month

cs2 <- as.data.table(read_parquet(IN_FILE))

# Step 1: Restrict to wearables, exclude StatTrak and Souvenir
n_raw_items <- uniqueN(cs2$item_name)
n_raw_base  <- uniqueN(cs2$base_item)

cs2 <- cs2[item_category %in% WEARABLE_CATS & !is_stattrak & !is_souvenir]

n_wear_items <- uniqueN(cs2$item_name)
n_wear_base  <- uniqueN(cs2$base_item)

# Step 2: Liquidity filter - drop any base_item with < 2 trades in any month
cs2[, month := format(date, "%Y-%m")]
monthly  <- cs2[, .(monthly_qty = sum(quantity)), by = .(base_item, month)]
illiquid <- monthly[monthly_qty < LIQUIDITY_MIN, unique(base_item)]
cs2      <- cs2[!base_item %in% illiquid]
```

```
cs2[, month := NULL]
n_liquid_base <- uniqueN(cs2$base_item)

# Save filter funnel for appendix table
filter_stats <- data.table(
  Stage = c("Raw items (all categories, all variants)",
            "After wearable categories + StatTrak/Souvenir exclusion",
            "After liquidity filter (>=2 trades/month in every month)"),
  Item_variants = c(n_raw_items, n_wear_items, NA_integer_),
  Base_items    = c(n_raw_base,  n_wear_base,  n_liquid_base)
)
fwrite(filter_stats, here("data", "processed", "filter_stats.csv"))

# Step 3: Value-weighted daily index
# I_t = sum(price_it * qty_it) / sum(qty_it)
index <- cs2[, .(
  index_level  = sum(trading_value) / sum(quantity),
  n_items      = uniqueN(base_item),
  total_volume = sum(quantity)
), by = date]
setorder(index, date)

write_parquet(index, OUT_FILE)
```

Step 3 — Exploratory Data Analysis (`src/03_eda.R`)

Loads the CS2 index and retrieves S&P 500, Bitcoin, and Gold prices via `quantmod`. Aligns all four series on common trading days, computes log returns, and produces descriptive statistics, a correlation matrix, ADF stationarity tests, and all EDA plots. Saves aligned log returns to `data/processed/returns.parquet` for downstream scripts.

Listing 3: Step 3: EDA — alignment, log returns, statistics, and plots

```
library(data.table); library(arrow); library(here)
library(ggplot2); library(patchwork); library(quantmod); library(tseries)

DATE_FROM <- "2013-08-01"; DATE_TO <- "2024-06-15"

# Load CS2 index
cs2 <- as.data.table(read_parquet(here("data","processed","cs2_index.parquet")))
cs2 <- cs2[, .(date, cs2 = index_level)]

# Download comparison assets
getSymbols(c("^GSPC","BTC-USD","GC=F"), from=DATE_FROM, to=DATE_TO,
           auto.assign=TRUE, warnings=FALSE)
sp500 <- data.table(date=as.Date(index(GSPC)),      sp500=as.numeric(Ad(GSPC)))
btc   <- data.table(date=as.Date(index(`BTC-USD`)), btc  =as.numeric(Ad(`BTC-USD`)))
gold  <- data.table(date=as.Date(index(`GC=F`)),    gold =as.numeric(Cl(`GC=F`)))

# Inner join on common trading days
dt <- Reduce(function(a,b) merge(a,b,by="date"), list(cs2,sp500,btc,gold))
```

```r
setorder(dt, date)

# Log returns
for (col in c("cs2","sp500","btc","gold"))
  dt[, paste0("r_",col) := c(NA_real_, diff(log(get(col))))]
ret <- dt[!is.na(r_cs2)]

# Descriptive statistics
desc_stats <- function(x) list(
  n=sum(!is.na(x)), mean=mean(x,na.rm=TRUE), sd=sd(x,na.rm=TRUE),
  min=min(x,na.rm=TRUE), max=max(x,na.rm=TRUE),
  skewness=mean(((x-mean(x,na.rm=TRUE))/sd(x,na.rm=TRUE))^3,na.rm=TRUE),
  kurtosis=mean(((x-mean(x,na.rm=TRUE))/sd(x,na.rm=TRUE))^4,na.rm=TRUE)
    -3
)
stats <- rbindlist(lapply(
  c(cs2="r_cs2",`S&P 500`="r_sp500",Bitcoin="r_btc",Gold="r_gold"),
  function(col) as.data.table(desc_stats(ret[[col]]))), idcol="asset")
print(stats)

# Correlation matrix
cor_mat <- cor(ret[,.(r_cs2,r_sp500,r_btc,r_gold)], use="complete.obs")
rownames(cor_mat) <- colnames(cor_mat) <- c("CS2","S&P 500","Bitcoin","
   Gold")

# ADF stationarity tests
for (col in c("r_cs2","r_sp500","r_btc","r_gold")) {
  adf <- adf.test(na.omit(ret[[col]]))
  message(sprintf("%-10s  stat=%6.3f  p=%.4f", col, adf$statistic, adf$p
    .value))
}

theme_set(theme_minimal(base_size=11))

# Figure 1: Normalised price levels (log scale)
levels_long <- melt(dt[, .(date,
  CS2=cs2/cs2[1]*100, `S&P 500`=sp500/sp500[1]*100,
  Bitcoin=btc/btc[1]*100, Gold=gold/gold[1]*100)],
  id.vars="date", variable.name="Asset", value.name="value")
p_levels <- ggplot(levels_long, aes(date,value,colour=Asset)) +
  geom_line(linewidth=0.5) + scale_y_log10() +
  labs(title="Normalised Price Levels (base=100, log scale)",
       x=NULL, y="Index (log scale)", colour=NULL) +
  theme(legend.position="bottom")
ggsave(here("images","eda","levels.png"), p_levels, width=10, height=4,
   dpi=150)

# Figure 2: Daily log returns
returns_long <- melt(ret[,.(date,CS2=r_cs2,`S&P 500`=r_sp500,
                            Bitcoin=r_btc,Gold=r_gold)],
  id.vars="date", variable.name="Asset", value.name="return")
p_returns <- ggplot(returns_long, aes(date,return)) +
  geom_line(linewidth=0.3, colour="steelblue") +
  facet_wrap(~Asset, ncol=1, scales="free_y") +
  labs(title="Daily Log Returns", x=NULL, y="Log Return")
ggsave(here("images","eda","returns.png"), p_returns, width=10, height
   =8, dpi=150)

```

```r
# Figure 3: Correlation heatmap
cor_long <- as.data.table(as.table(cor_mat))
setnames(cor_long, c("Asset1","Asset2","correlation"))
p_cor <- ggplot(cor_long, aes(Asset1,Asset2,fill=correlation)) +
  geom_tile(colour="white") + geom_text(aes(label=round(correlation,2)), size=3.5)+
  scale_fill_gradient2(low="#d73027",mid="white",high="#1a9850",midpoint=0,
                       limits=c(-1,1)) +
  labs(title="Correlation Matrix", x=NULL, y=NULL, fill="r") +
  theme(axis.text.x=element_text(angle=45,hjust=1))
ggsave(here("images","eda","correlation.png"), p_cor, width=5, height=4, dpi=150)

# Figure 4: ACF and PACF of log returns
png(here("images","eda","acf.png"), width=1200, height=800, res=120)
par(mfrow=c(2,4), mar=c(4,4,3,1))
for (col in c("r_cs2","r_sp500","r_btc","r_gold")) {
  label <- switch(col,r_cs2="CS2",r_sp500="S&P 500",r_btc="Bitcoin",r_gold="Gold")
  acf(na.omit(ret[[col]]),  main=paste("ACF -",  label), lag.max=30)
  pacf(na.omit(ret[[col]]), main=paste("PACF -", label), lag.max=30)
}
dev.off()

# Figure 5: Return distributions with normal overlay
norm_curves <- returns_long[, {
  x_seq <- seq(min(return,na.rm=TRUE), max(return,na.rm=TRUE), length.out=300)
  .(x=x_seq, y=dnorm(x_seq, mean(return,na.rm=TRUE), sd(return,na.rm=TRUE)))
}, by=Asset]
p_dist <- ggplot(returns_long, aes(x=return)) +
  geom_histogram(aes(y=after_stat(density)), bins=60,
                 fill="steelblue", colour="white", linewidth=0.1) +
  geom_line(data=norm_curves, aes(x=x,y=y),
            colour="red", linewidth=0.7, inherit.aes=FALSE) +
  facet_wrap(~Asset, scales="free", ncol=2) +
  labs(x="Log return", y="Density",
       title="Return Distributions with Normal Overlay (red)") +
  theme_minimal(base_size=10)
ggsave(here("images","eda","distributions.png"), p_dist, width=8, height=5, dpi=150)

# Save aligned returns for downstream scripts
write_parquet(ret, here("data","processed","returns.parquet"))
```

Step 4 — GARCH Volatility Modelling (`src/04_garch.R`)

Fits a GARCH(1,1) model with student-t innovations to each of the four return series using the `rugarch` package. Extracts and prints the ARCH coefficient $\hat{\alpha}$, GARCH coefficient $\hat{\beta}$, persistence, and annualised unconditional volatility. Produces a conditional volatility plot and a standardised residuals Q-Q diagnostic.

Listing 4: Step 4: GARCH(1,1) estimation and volatility diagnostics

```r
library(data.table); library(arrow); library(here)
```

```r
library(rugarch); library(ggplot2)

ret <- as.data.table(read_parquet(here("data","processed","returns.parquet")))

ASSETS <- c(CS2="r_cs2", `S&P 500`="r_sp500", Bitcoin="r_btc", Gold="r_gold")

# GARCH(1,1) specification with student-t innovations
spec <- ugarchspec(
  variance.model     = list(model="sGARCH", garchOrder=c(1,1)),
  mean.model         = list(armaOrder=c(0,0), include.mean=TRUE),
  distribution.model = "std"
)

fits <- list(); params <- list()

for (label in names(ASSETS)) {
  x   <- na.omit(ret[[ASSETS[[label]]]])
  fit <- ugarchfit(spec=spec, data=x, solver="hybrid")
  fits[[label]] <- fit

  cf    <- coef(fit)
  alpha <- cf["alpha1"]; beta <- cf["beta1"]; omega <- cf["omega"]
  pers  <- alpha + beta
  uncond_vol_ann <- sqrt(omega / (1 - pers)) * sqrt(252) * 100

  params[[label]] <- data.table(asset=label, omega=omega, alpha=alpha,
    beta=beta, persistence=pers, uncond_vol_ann=uncond_vol_ann)

  message(sprintf("%s: alpha=%.4f  beta=%.4f  persistence=%.4f  vol=%.2f %%",
                  label, alpha, beta, pers, uncond_vol_ann))
}

results <- rbindlist(params)
print(results[, .(asset, alpha=round(alpha,4), beta=round(beta,4),
                  persistence=round(persistence,4),
                  uncond_vol_ann=round(uncond_vol_ann,2))])

# Figure: Conditional volatility (annualised)
vol_dt <- rbindlist(lapply(names(fits), function(label) data.table(
  date  = ret$date[!is.na(ret[[ASSETS[[label]]]])],
  asset = label,
  sigma = as.numeric(sigma(fits[[label]])) * sqrt(252) * 100
)))
vol_dt[, asset := factor(asset, levels=c("CS2","S&P 500","Bitcoin","Gold"))]
p_vol <- ggplot(vol_dt, aes(date,sigma)) +
  geom_line(linewidth=0.4, colour="steelblue") +
  facet_wrap(~asset, ncol=1, scales="free_y") +
  labs(title="GARCH(1,1) Conditional Volatility (Annualised, %)",
       x=NULL, y="Conditional Volatility (%)")
ggsave(here("images","garch","conditional_volatility.png"),
       p_vol, width=10, height=9, dpi=150)

# Figure: Standardised residuals Q-Q plots (motivation for student-t)
png(here("images","garch","residual_qq.png"), width=1200, height=800,
```

```
    res=150)
par(mfrow=c(2,2), mar=c(4,4,3,1))
for (label in names(fits)) {
  std_resid <- as.numeric(residuals(fits[[label]], standardize=TRUE))
  qqnorm(std_resid, main=paste("Std. Residuals Q-Q -", label),
         cex.main=0.85, pch=16, cex=0.4, col="steelblue")
  qqline(std_resid, col="red", lwd=1.5)
}
dev.off()
```

Step 5 — VAR and Granger Causality (`src/05_var_granger.R`)

Estimates a VAR(p) model on the joint system of four log return series using the `vars` package. Selects the lag order by BIC via `VARselect`, runs joint Granger causality F-tests via `causality()`, and saves residual diagnostics. The fitted VAR object is saved to `data/processed/var_model.rds` for potential reuse.

Listing 5: Step 5: VAR lag selection, Granger causality, and residual diagnostics

```
library(data.table); library(arrow); library(here); library(vars)

ret      <- as.data.table(read_parquet(here("data","processed","returns.
    parquet")))
var_data <- na.omit(ret[, .(r_cs2, r_sp500, r_btc, r_gold)])

# Lag selection: AIC=9, BIC=4 -> use BIC for parsimony
lag_select <- VARselect(var_data, lag.max=10, type="const")
print(lag_select$selection)

lag_table <- as.data.table(t(lag_select$criteria), keep.rownames="Lag")
lag_table[, Lag := as.integer(Lag)]
fwrite(lag_table, here("data","processed","var_lag_selection.csv"))

p <- lag_select$selection["SC(n)"]   # BIC-selected lag order

# Fit VAR(p)
var_model <- VAR(var_data, p=p, type="const")
summary(var_model)

# Granger causality: each variable tested against the entire remaining
    system
tests <- list(
  "Bitcoin  -> system" = "r_btc",
  "S&P 500  -> system" = "r_sp500",
  "Gold     -> system" = "r_gold",
  "CS2      -> system" = "r_cs2"
)

granger_results <- rbindlist(lapply(names(tests), function(name) {
  res   <- causality(var_model, cause=tests[[name]])
  pval  <- res$Granger$p.value
  fstat <- res$Granger$statistic
  message(sprintf("%-25s  F=%6.3f  p=%.4f", name, fstat, pval))
  data.table(direction=name, F_stat=round(fstat,3), p_value=round(pval
      ,4))
}))
print(granger_results)
```

```r

saveRDS(var_model, here("data","processed","var_model.rds"))

# Ljung-Box residual autocorrelation tests (lag=10)
resid_mat    <- residuals(var_model)
asset_labels <- c(r_cs2="CS2",r_sp500="S&P 500",r_btc="Bitcoin",r_gold="
    Gold")

lb_results <- rbindlist(lapply(colnames(resid_mat), function(col) {
  test <- Box.test(resid_mat[, col], lag=10, type="Ljung-Box")
  data.table(Asset=asset_labels[col], Q_stat=round(test$statistic,3),
             p_value=round(test$p.value,4), Df=test$parameter)
}))
print(lb_results)
fwrite(lb_results, here("data","processed","var_ljungbox.csv"))
```

Declaration of Authorship

We hereby declare that the work submitted is our own and that we have used no sources other than those explicitly acknowledged in the bibliography. All passages taken verbatim or substantially paraphrased from published or unpublished sources have been clearly identified as such. This work has not been submitted, in the same or substantially similar form, to any other examination authority.

Lucerne, 22 May 2026

Emanuel Lemma

Lucerne, 22 May 2026

Jakub Holzmann